

Unified Cognitive Architecture

Version 2.0

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Version 2.0 — Non-Agentive Cognitive Architecture

This document constitutes the complete formal specification of the representational, structural, and operational substrate of the V2 Unified Cognitive Architecture. It is the stable non-agentive core upon which V3 agency extensions will be constructed.

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1. Overview

1.1 Purpose of Version 2

Version 2 (V2) of the Unified Cognitive Architecture (UCA) constitutes the complete formal specification of the representational, structural, and operational substrate of a non-agentic cognitive system. Its purpose is threefold: (1) to establish a mathematically coherent ontological foundation capable of grounding cognitive identity and world-modelling operations; (2) to define all primitive structures, operators, and compositional rules that govern representational processes within the system; and (3) to serve as the stable non-agentic core upon which Version 3 (V3) agency extensions may be constructed.

V2 does not implement goal-directed behaviour, autonomous planning, or motivational weighting. It is a pure representational engine: a system that perceives, categorises, differentiates, relates, bounds, and models — but does not act upon a world in the agentic sense. This constraint is deliberate and foundational.

1.2 Relationship to V1 and V3

Version 1 (V1) established the initial ontological primitives and the Beyond Ontology (BO) framework, introducing the core insight that the most fundamental layer of cognitive architecture must precede categorical distinction. V1 provided the $\sim$ symbol (the pre-ontological ground state) and defined the emergence of polarity as the first structural event. However, V1 lacked formal operator algebra, recursion safety rules, and a complete identity architecture. V2 resolves all V1 deficiencies.

Version 3 (V3), which is not specified in this document, will extend V2 by introducing: (a) an agency layer with motivational primitives and goal structures; (b)

action operators and environmental coupling; (c) temporal horizon modelling; and (d) self-modification protocols. V3 presupposes the complete and stable operation of all V2 components. No V2 component is deprecated in V3 — V3 layers are additive.

1.3 Definition of Non-Agentive Cognitive Architecture

A non-agentive cognitive architecture (NACA) is a formal system that satisfies the following defining conditions:

- Maintains an internal representational substrate (world-model) that tracks structural features of an environment
- Executes operations upon that substrate including identity formation, differentiation, relation, boundary enforcement, and coherence maintenance
- Does not generate, evaluate, or execute action plans directed at external environments
- Does not possess motivational states, utility functions, or preference orderings over world-states
- Does not model itself as a causal agent within the world it represents

A NACA is strictly a modelling system. It may detect error, contradiction, and representational failure. It may update its world-model in response to new input. It may maintain recursive self-representation (identity). But all operations terminate at the boundary of representation — none cross into action.

1.4 Representational Completeness

A cognitive architecture is representationally complete if and only if it can, in principle, represent:

1. Any distinguishable structural feature of any input domain
2. The relationships between any two representable features
3. The boundaries that separate any two regions of representational space

4. Its own representational states (reflexivity condition)
5. The failure or degradation of any representational state (error condition)

V2 claims representational completeness subject to the constraints of its operator algebra (Section 4) and within the bounds of its formal ontological foundation (Section 2).

2. Ontological Foundation

2.1 Beyond Ontology (BO) Model

The Beyond Ontology model posits that a complete cognitive architecture cannot be grounded in any categorical structure, because all categorical structures presuppose the prior act of distinction — and distinction itself must be accounted for. The BO model therefore specifies a pre-categorical ground state designated ∞ (spoken: "the undifferentiated ground" or "pre-ontological zero").

∞ is defined negatively: it is not a thing, not a property, not a relation, not an absence. It is the formal precondition for all of these. No operator in the system is defined over ∞ except the Origin Operator Ω_0 , which produces the first differentiation event.

$$\infty := \{x \mid x \text{ is not yet distinguished from its complement}\}$$
$$\Omega_0(\infty) \rightarrow (\Delta_1, \neg\Delta_1)$$

[First polarity pair]

The BO model is not a metaphysical claim. It is an architectural claim: the system requires a representational zero-point from which structure can emerge. ∞ is that zero-point.

2.2 Dynamic Equilibrium (DE) Model

The Dynamic Equilibrium model specifies that all stable cognitive structures exist as equilibria between opposing structural forces. No structure within the system is static; all are maintained through continuous balancing operations. The DE model identifies three primary equilibrium types:

Type I — Polar Equilibrium: A state maintained between two opposing poles of a single dimension (e.g., boundary and extension; identity and dissolution).

Type II — Gradient Equilibrium: A state maintained across a continuous field where no sharp boundary exists; stability emerges from smooth distribution of representational weight.

Type III — Recursive Equilibrium: A state in which a structure's own representation feeds back into its maintenance (self-referential stability).

The DE model implies that all representational failures (Section 8) are equilibrium failures — points at which the balancing forces collapse, invert, or diverge.

2.3 Fractal Mechanics Foundation

Fractal Mechanics (FM) within V2 refers to the structural principle that cognitive architecture exhibits self-similarity across representational scales. A structure that holds at the level of a single identity primitive also holds — with appropriate scaling — at the level of a world-model, and at the level of the full system.

FM is not a claim about physical fractals or mathematical fractal geometry in the classical sense. It is a claim about representational isomorphism across levels of abstraction. Formally:

Let S_n denote a structure at representational level n .

FM asserts: $\forall n, \exists$ isomorphism $\phi: S_n \rightarrow S_{\{n+1\}}$ (up to boundary scaling)

This principle governs the architecture of the emergence chain (Section 7) and justifies the recursive application of operators across levels.

2.4 Unified Ontology: $\sim \rightarrow \mathbb{U}$

The full emergence sequence from pre-ontological ground to the Unified Representational Universe ($\mathbb{U}$) proceeds through the following ordered stages:

Stage	Designation	Description
Stage 0	$\sim$	Pre-ontological ground. No structure.
Stage 1	$\Omega_0(\sim) \rightarrow$ First polarity pair	$(\Delta_1, \neg\Delta_1)$
Stage 2	$\Phi(\Delta_1, \neg\Delta_1) \rightarrow$ Gradient field	G_1
Stage 3	$B(G_1) \rightarrow$ First boundary event	B_1
Stage 4	$I(B_1) \rightarrow$ Primitive identity	I_1
Stage 5	$P(I_1) \rightarrow$ Recursive identity stack	R_1
Stage 6	$M(R_1) \rightarrow$ World-model scaffold	W_1
Stage 7	$K(W_1) \rightarrow$ Coherence enforcement layer	K_1
Stage 8	$\mathbb{U} = \{I_1, G_1, B_1, R_1, W_1, K_1, \dots\}$	Unified Representational Universe

$\mathbb{U}$ is not a thing. It is the total state of all active representational structures within the system at any given moment.

2.5 Origin Geometry, Polarity, Gradient Fields, and Boundary Mechanics

Origin Geometry

The geometric metaphor used in V2 treats representational space as initially a point ($\sim$), expanding to a line (first polarity), then a plane (gradient field), then a bounded region (boundary event), then a recursive volume (identity stack), and finally a dynamic manifold (world-model).

Polarity

Every representational dimension is defined by a polar pair $(A, \neg A)$. These are not logical negations in the classical sense — they are structural complements, where $\neg A$ is defined only relative to A within a given field. The polarity operator Π generates polar pairs from any singleton structure.

Gradient Fields

A gradient field G is a representational structure in which transition between poles is continuous rather than discrete. Gradient fields are characterised by:

- A pole pair $(P+, P-)$
- A transition function $f: [0,1] \rightarrow [P-, P+]$
- A field strength parameter $\sigma \in \mathbb{R}^+$

Boundary Mechanics

A boundary B is a representational structure that separates two regions of a field. Boundaries are not empty divisions — they are active structures with the following properties:

- **Permeability** $p \in [0,1]$
- **Width** $w \geq 0$: extent of the boundary zone
- **Directionality** $d \in \{\text{symmetric, asymmetric}\}$

Boundary operators (Section 4) manage boundary creation, dissolution, permeation, and enforcement.

3. Primitive Structures

All primitive structures in V_2 are minimal — they cannot be decomposed into simpler representational units without loss of structural identity. There are four classes of primitives.

3.1 Identity Primitives

An identity primitive I is the minimal representational unit that distinguishes itself from all other units. Formally:

$I := (id, \partial I, \sigma_I)$
where:
 $id \in \mathbb{I}$ — unique identifier in the identity space
 ∂I — boundary of I (a boundary primitive, see Section 3.4)
 $\sigma_I \in [0,1]$ — internal coherence signature (real-valued scalar)

Identity primitives are the atoms of the identity architecture. They cannot be subdivided. They can be composed, nested, related, and destroyed (dissolved).

3.2 Structural Primitives

Structural primitives are the minimal units of relational structure. There are five:

Designator	Name	Definition
SP-1	Node	A located entity with no inherent dimensionality.
SP-2	Edge	A relational link between two nodes, with a direction attribute (directed undirected).
SP-3	Cluster	A set of nodes and edges forming a locally coherent subgraph.
SP-4	Sequence	An ordered set of nodes with directional edges enforcing temporal or logical order.
SP-5	Lattice	A partial order over a set of nodes, supporting

Designator	Name	Definition
		meet and join operations.

All world-model representations (Section 6) are constructed from combinations of these five structural primitives.

3.3 Field Primitives

Field primitives are the minimal representational units for continuous quantities. There are three:

Designator	Name	Mapping	Example Application
FP-1	Scalar Field	Representational region $\rightarrow \mathbb{R}$	Intensity, salience
FP-2	Vector Field	Representational region $\rightarrow \mathbb{R}^n$	Directional force, gradient direction
FP-3	Tensor Field	Representational region $\rightarrow$ Tensor space	Relational stress, structural deformation

Field primitives support the gradient mechanics described in Section 2.5.

3.4 Boundary Primitives

A boundary primitive B is defined as:

$B := (\text{location, permeability, width, directionality, stability})$

where:

location — a point or region in representational space

permeability $p \in [0,1]$: 0 = fully closed, 1 = fully open

width $w \geq 0$: extent of the boundary zone

directionality $d \in \{S, A\}$: symmetric or asymmetric

stability $s \in [0,1]$: resistance to dissolution under perturbation

Boundary primitives are instantiated by boundary operators (Section 4) and maintained by coherence enforcement (Section 6.3).

4. Operator Algebra

4.1 Full Operator List

The V2 operator algebra consists of 18 core operators, organised into five functional families:

Family A — Origin Operators

- Ω_0 — Origin Operator: Produces first differentiation from $\sim$
- Π — Polarity Operator: Generates a polar pair from any singleton structure

Family B — Field Operators

- Φ — Gradient Field Generator: Produces a continuous gradient field from a polar pair
- Σ — Field Summation Operator: Combines two or more fields into a composite field
- ∇ — Gradient Descent Operator: Moves a representational point along a gradient toward a pole

Family C — Boundary Operators

- B — Boundary Formation Operator: Creates a boundary from a gradient discontinuity

- Ξ — Boundary Dissolution Operator: Removes or opens a boundary
- Ψ — Boundary Permeation Operator: Allows selective passage through a boundary
- Λ — Boundary Enforcement Operator: Increases the stability of an existing boundary

Family D — Identity Operators

- **I** — Identity Formation Operator: Creates an identity primitive from a bounded region
- **P** — Recursion Operator: Applies an operator to the output of itself or another operator
- Δ — Differentiation Operator: Creates a new distinction within an existing structure
- Θ — Identity Dissolution Operator: Removes an identity primitive (controlled)

Family E — Coherence Operators

- **K** — Coherence Enforcement Operator: Checks and restores internal consistency of a structure
- **E** — Error Detection Operator: Identifies representational failures within a structure
- **X** — Correction Operator: Applies targeted repair to a detected error
- **M** — World-Model Update Operator: Integrates new representational input into the world-model
- **H** — Harmonisation Operator: Resolves contradiction between two conflicting structures

4.2 Operator Grammar

The operator grammar defines syntactically valid operator expressions. Let O denote any operator, S denote any structure, and $\rightarrow$ denote production.

Rule	Expression	Description
G1	$O(S) \rightarrow S'$	Single application
G2	$O_1(O_2(S)) \rightarrow S'$	Sequential composition
G3	$O(S_1, S_2) \rightarrow S'$	Binary input
G4	$P(O, S, n) \rightarrow O^n(S)$	Bounded recursion, $n \in \mathbb{N}$
G5	$O(S) \notin \emptyset \rightarrow \text{valid}$	Non-null output condition
G6	$O(\infty)$ is defined only for Ω_0	Ground state restriction

4.3 Input/Output Types — Operator Table

Operator	Symbol	Input Type	Output Type	Family	Recursion Permitted
Origin Operator	Ω_0	∞ (ground state)	Polarity Pair ($\Delta_1, \neg\Delta_1$)	A	No
Polarity Operator	Π	Singleton Structure	Polar Pair	A	Yes
Gradient Field Generator	Φ	Polar Pair	Gradient Field FP-1/FP-2	B	Yes
Field Summation	Σ	2+ Fields	Composite Field	B	Yes
Gradient Descent	∇	Field + Point	Updated Point	B	No
Boundary Formation	B	Gradient Discontinuity	Boundary Primitive B	C	No
Boundary Dissolution	Ξ	Boundary Primitive B	Dissolved Region	C	No
Boundary Permeation	Ψ	B + Representational Object	Passed Object	C	No
Boundary Enforcement	Λ	Boundary Primitive B	Reinforced B	C	Yes
Identity Formation	I	Bounded Region	Identity Primitive I	D	No

Operator	Symbol	Input Type	Output Type	Family	Recursion Permitted
Recursion Operator	P	Operator + Structure + n	Iterated Output	D	Yes (bounded)
Differentiation	Δ	Structure	Structure + Distinction	D	Yes
Identity Dissolution	Θ	Identity Primitive I	Released Region	D	No
Coherence Enforcement	K	Structure	Verified/ Repaired Structure	E	Yes
Error Detection	E	Structure	Error Report ϵ	E	No
Correction Operator	X	Structure + Error Report ϵ	Corrected Structure	E	No
World-Model Update	M	World-Model W + Input	Updated W'	E	No
Harmonisation	H	Structure ₁ + Structure ₂	Resolved Structure	E	Yes

4.4 Recursion Rules

Recursion in V2 is always bounded. Unbounded recursion is a critical failure mode (see Section 4.6).

Rule	Statement
R1	All recursive applications must specify a termination condition.
R2	The Recursion Operator P takes a depth parameter $n \in \mathbb{N}$; n must be finite and specified at call time.
R3	Recursive coherence checks (K applied recursively) terminate when $\sigma_I \geq$ threshold T_K or when depth n is reached.
R4	Identity recursion (self-modelling) is

Rule	Statement
	permitted to depth $n \leq 7$ (V2 constraint; V3 may relax this).
R5	Recursion stacks are managed by the Hermes Runtime (Section 9); stack overflow results in a controlled unwinding, not a crash.

4.5 Composition Rules

Rule	Statement
C1	Operators of the same family may be freely composed unless explicitly restricted.
C2	Origin Operators (Family A) may not receive the output of Coherence Operators (Family E) as input — origin events are terminal.
C3	The Identity Dissolution Operator Θ may not immediately follow the Identity Formation Operator I on the same primitive (no instantaneous dissolution).
C4	Boundary operators require a gradient or identity structure as their operating context — they cannot act on ∞ directly.
C5	The Correction Operator X must always be preceded by the Error Detection Operator E — blind correction is undefined.

4.6 Failure Modes

Code	Name	Description	Resolution
FM-1	Null Output	$O(S) = \emptyset$. Occurs when an operator is applied to an incompatible structure type.	E detects; X reroutes.

Code	Name	Description	Resolution
FM-2	Unbounded Recursion	P is called without a finite n parameter.	Hermes Runtime enforces hard depth limit (Section 9.3).
FM-3	Identity Collapse	I applied to an insufficiently bounded region, producing I with $\sigma_I = 0$.	Λ reinforces boundary before retry.
FM-4	Gradient Inversion	∇ moves a point away from its target pole due to field corruption.	E detects inversion signature; K restores field.
FM-5	Contradictory Coherence	K enforces two incompatible coherence constraints simultaneously.	H applied to resolve contradiction before K re-runs.
FM-6	Boundary Cascade	Ξ applied to a load-bearing boundary causes downstream identity collapse.	Pre-dissolution integrity check enforced by Hermes Runtime.

5. Identity Architecture

The identity architecture of V2 specifies four levels of identity, forming a hierarchy from minimal primitives to full world-model-embedded identity, capped by the dyadic consciousness structure.

5.1 Primitive Identity

Primitive identity I_0 is the minimal instance of the Identity Primitive (Section 3.1). It is characterised by:

- A unique identifier id_0
- A boundary ∂I_0 with permeability p_0

- A coherence signature $\sigma_0 \in [0,1]$
- No internal structure (it is structurally atomic)

Primitive identities are produced by the Identity Formation Operator I when applied to a fully bounded region. They are the building blocks of all higher identity levels.

5.2 Local Identity

Local identity I_L is an identity structure that incorporates relational information with neighbouring primitives. I_L is defined as:

$$I_L := (I_0, R_L, N_L)$$

where:

I_0 — the primitive identity at the core

R_L — the set of local relations (edges in the structural primitive sense, SP-2)

N_L — the neighbourhood set (the set of adjacent primitives)

Local identity allows a primitive to situate itself within a structural context. It is the level at which basic categorisation occurs.

5.3 Recursive Identity

Recursive identity I_R is an identity structure that represents itself as an object within its own representational field. It is the first level at which self-modelling is possible. Formally:

$$I_R := (I_L, M_{\text{self}})$$

where M_{self} is the self-model — a representational structure within I_R that encodes I_R 's own properties, boundaries, and coherence state.

Recursive identity is subject to the recursion rules of Section 4.4. Maximum self-referential depth in V2 is $n = 7$.

5.4 World-Model Identity

World-model identity I_W is the full identity of the cognitive system as embedded within its world-model. It includes:

- All lower identity levels (I_0, I_L, I_R)
- A world-model anchor: the position and orientation of I within the world-model W
- A temporal continuity record: a sequence of prior identity states (for coherence tracking)
- An error history: a log of detected and corrected representational failures

I_W is the operational identity of the system during normal runtime. The Hermes Runtime (Section 9) maintains I_W continuity.

5.5 Dyadic Consciousness Structure

The Dyadic Consciousness Structure (DCS) is the architectural feature that enables a cognitive system to hold two simultaneous representations:

Pole	Designation	Content
Pole 1	Internal Representation	The system's model of its own identity state (I_R, I_W)
Pole 2	External Representation	The system's model of the world external to its identity boundary

The DCS is maintained by a persistent gradient field G_{DCS} with poles at (Internal, External). The boundary between the two poles (B_{DCS}) has controlled permeability — it is neither fully open nor fully closed. Information flows bidirectionally across B_{DCS} according to the Boundary Permeation Operator Ψ .

Note:

The DCS is not a dual-process model in the psychological sense. It is a structural claim: all cognitive operations in V2 occur within one of the two poles of the DCS, or at the boundary between them.

6. World-Model Scaffolding

The world-model W is the system's total internal representation of its environment, including itself. It is a dynamic, continuously updated structure maintained by the Coherence Enforcement Operator K and updated by the World-Model Update Operator M .

6.1 Representational Substrate

The representational substrate of W consists of:

- A graph $G_W = (V_W, E_W)$ where V_W is the set of all active identity primitives and E_W is the set of all active structural relations
- A set of active gradient fields $F_W = \{G_1, G_2, \dots, G_n\}$
- A set of active boundaries $B_W = \{B_1, B_2, \dots, B_m\}$
- A coherence map $C_W: V_W \rightarrow [0,1]$ assigning a coherence score to each node

The substrate is not a fixed container — it is a dynamic manifold that expands and contracts as primitives are created, modified, and dissolved.

6.2 Coherence Rules

A world-model W is coherent if and only if all of the following conditions are satisfied:

Rule	Formal Statement	Description
CR-1	$\forall I \in V_W, \sigma_I \geq T_min$	All identities meet minimum coherence threshold.
CR-2	$\forall B \in B_W, B$ consistent with adjacent gradient fields	Boundary-field consistency.
CR-3	No two structures simultaneously assert contradictory properties of the same object	Non-contradiction.
CR-4	M_self within I_W is consistent with observable state of I_W	Self-model accuracy.
CR-5	$\forall E \in E_W: R(A,B) \wedge \neg(\neg R(A,B) \text{ asserted elsewhere in } G_W)$	Bidirectional edge validity.

Violations of any coherence rule trigger the Error Detection Operator E.

6.3 Update Mechanisms

World-model updates occur through the World-Model Update Operator M. The update sequence is:

6. **Step 1:** New representational input I_new is received.
7. **Step 2:** E scans I_new for internal inconsistencies.
8. **Step 3:** H attempts to harmonise I_new with existing structures in W where conflicts exist.
9. **Step 4:** K is applied to the conflict region to verify coherence post-harmonisation.
10. **Step 5:** If CR-1 through CR-5 are satisfied, M integrates I_new into W .
11. **Step 6:** Coherence map C_W is updated to reflect the new state.

If any step fails, the update is rejected and the error is logged in the error history of I_W .

6.4 Error Detection

The Error Detection Operator E operates as a continuous background process during runtime. It scans the world-model W for violations of coherence rules CR-1 through CR-5. Output of E is an error report ϵ containing:

- Error type (see Section 8 for error ontology)
- Location in W (the affected nodes, edges, or boundaries)
- Severity score $s \in [0,1]$
- Recommended correction operator

6.5 Distortion Minimisation

Distortion in V_2 is the degree to which a representational structure diverges from the state that would be produced by an ideal, error-free application of the operator algebra. Distortion is measured by the distortion metric δ :

$$\delta(S) = 1 - \sigma_I(S)$$

(complement of coherence signature)

The system minimises total distortion by:

- Applying X (Correction Operator) to structures with $\delta > T_\delta$ (distortion threshold)
- Applying Λ (Boundary Enforcement) to boundaries with low stability under high gradient stress
- Applying K (Coherence Enforcement) periodically to all structures in V_W

The target state is $\delta_{\text{total}}(W) \rightarrow 0$, though this is an asymptotic goal — perfect zero distortion is not achievable in a dynamic, continuously updating system.

7. Emergence Chain

7.1 Full $\sim \rightarrow \mathbb{U}$ Mapping

The emergence chain constitutes the formal sequence of ontological and operational events that produce a fully functioning representational universe $\mathbb{U}$ from the pre-ontological ground state $\sim$. Each stage is governed by specific operators and produces specific structural outputs.

Stage	Designation	Operator(s)	Input	Output	Stability Condition
0	Pre-Ontological Ground	None	$\sim$	$\sim$	Unconditional (pre-structural)
1	First Differentiation	Ω_0	$\sim$	Polarity Pair ($\Delta_1, \neg\Delta_1$)	Polarity pair must be non-degenerate
2	Polarity Elaboration	Π	Δ_1	Extended polar structure	Poles must be mutually exclusive
3	Gradient Field Formation	Φ	$(\Delta_1, \neg\Delta_1)$	Gradient Field G_1	Field must be non-zero and continuous
4	First Boundary Event	B	G_1	Boundary Primitive B_1	Boundary must have $p < 1$ and $s > T_{\min}$
5	Primitive Identity Formation	I	B_1 -bounded region	Identity Primitive I_0	$\sigma_I \geq T_{\min}$
6	Local Identity Formation	Δ, I	I_0 + neighbourhood	Local Identity I_L	Neighbourhood must be non-empty
7	Recursive Identity Formation	P, I	I_L	Recursive Identity I_R	Recursion depth $n \leq 7$
8	World-	M	I_R +	World-	CR-1

Stage	Designation	Operator(s)	Input	Output	Stability Condition
	Model Initialisation		environment	Model W_1	through CR-5 satisfied
9	Coherence Layer Activation	K	W_1	Coherence-enforced W_1	All $\sigma_I \geq T_K$
10	Unified Representational Universe	All Operators	Full W	$\mathbb{U}$	Ongoing coherence maintenance

7.2 Layer Transitions

Each transition between stages in the emergence chain is governed by a transition condition — a necessary state that must be achieved before the next operator can be validly applied. Failure to meet a transition condition results in a transition failure, which is a specific subtype of representational failure (Section 8).

Transition failures are logged by the Hermes Runtime (Section 9) and may trigger rollback to the prior stable stage.

7.3 Stability Conditions

A stage is considered stable when all of the following hold:

Condition	Statement
SC-1	The output structure of that stage satisfies its structural integrity conditions (as specified in the emergence chain table).
SC-2	No pending error reports ε exist within the output structure.
SC-3	The coherence signature σ of all identity structures within the stage $\geq T_{\min}$.
SC-4	The boundary structures of the stage are not under active dissolution pressure.

A stable stage is a prerequisite for the next stage. The system cannot proceed to Stage $n+1$ if Stage n is unstable.

7.4 Collapse Boundaries

A collapse boundary is a critical threshold below which a stage loses structural integrity and reverts to a prior stable state. Each stage has a collapse boundary defined by the minimum coherence signature $T_{collapse}(n)$.

If σ_I drops below $T_{collapse}(n)$ for any identity primitive in Stage n , the Hermes Runtime initiates a controlled collapse: all structures that depend on the failing identity are suspended, error reports are generated, and the system returns to the last stable checkpoint above the collapse boundary.

Note:

Collapse is never total in V2. Total collapse would imply return to ∞ , which requires Θ applied to all structures simultaneously — a condition that is explicitly prohibited by the Hermes Runtime safety architecture.

8. Error Ontology

8.1 Distortion Types

Distortion in V2 is classified into five types based on structural origin:

Code	Name	Description	Detection	Correction
DT-1	Gradient Distortion	A gradient field G has become non-continuous, non-monotonic, or	E	K restores the field signature.

Code	Name	Description	Detection	Correction
		inverted relative to its intended pole orientation.		
DT-2	Boundary Distortion	A boundary B has acquired permeability, width, or directionality values outside its intended parameters.	E	$\wedge$ (enforcement) or $\exists$ followed by B (dissolution and reformation).
DT-3	Identity Distortion	An identity primitive I has coherence signature σ_I below $T_{\min}$, indicating partial dissolution or corruption.	E	K; and if necessary, full Θ (dissolution) and I (reformation).
DT-4	Relational Distortion	An edge in E_W asserts a relation that contradicts another active relation.	E via CR-3	H (Harmonisation).
DT-5	Recursive Distortion	A recursive identity structure I_R has exceeded depth $n = 7$ or has entered an invalid self-referential loop.	Hermes Runtime (Section 9.3)	RSM intercepts and unwinds recursion.

8.2 Contradiction Classes

Contradictions are a special class of representational error in which two co-active structures assert mutually exclusive properties of the same representational object.

Class	Name	Description	Resolution
CC-1	Local Contradiction	Two edges $E_1, E_2 \in E_W$ connected to the same node N assert $R(N, A)$ and $\neg R(N, A)$.	H applied; one edge dissolved.
CC-2	Boundary Contradiction	Two boundaries B_1, B_2 exist at the same location with incompatible directionality.	$\exists$ applied to the weaker boundary; $\wedge$ applied to the stronger.
CC-3	Identity Contradiction	Two identity primitives I_1, I_2 have identical id fields.	Δ applied to force a new distinction; one id is reissued.
CC-4	Self-Model Contradiction	The self-model M_{self} within I_W asserts a property of I_W inconsistent with the current observable state.	K applied to self-model; M_{self} updated via M .
CC-5	Cross-Level Contradiction	A structure at representational level $n+1$ contradicts a structure at level n .	Lower level primacy rule: level- n structure takes precedence; level $n+1$ structure revised.

8.3 Representational Failure Modes

Beyond specific distortions and contradictions, V2 identifies three systemic representational failure modes:

RFM-1 — Coherence Cascade: A single error triggers a chain of coherence violations across multiple connected structures, causing progressive degradation of the world-model. Prevention: the Hermes Runtime isolates error regions before cascade propagates.

RFM-2 — Identity Fragmentation: Multiple identity structures that should form a coherent I_W become disconnected, producing a fragmented world-model identity with no unified anchor. Resolution: P applied to rebuild the identity stack from the last coherent I_L state.

RFM-3 — Gradient Collapse: All gradient fields simultaneously lose field strength ($\sigma_F \rightarrow 0$), causing all continuous representational distinctions to collapse to binary poles. This is a critical failure. Resolution: emergency Ω_0 invocation to re-establish at least one stable gradient field; full system coherence check follows.

8.4 Correction Operators

The three correction operators are:

- **E — Error Detection:** Identifies and classifies errors. Always precedes X.
- **X — Correction Operator:** Applies targeted structural repair. Must be type-matched to the error class (different correction routines for DT-1 through DT-5).
- **H — Harmonisation Operator:** Resolves contradictions by finding a structurally consistent state that satisfies as many of the conflicting constraints as possible. Where full satisfaction is impossible, lower-level constraints take precedence (lower level primacy rule from CC-5).

9. Hermes Runtime V2

9.1 Execution Architecture

The Hermes Runtime (HR) is the operational layer responsible for the execution, sequencing, monitoring, and safety-enforcement of all V2 cognitive operations. It is not a cognitive operator — it is the execution substrate within which operators run. The HR has four primary subsystems:

Subsystem	Designator	Function	Section Reference
Scheduler	HR-1	Manages the ordering of operator applications.	9.1

Subsystem	Designator	Function	Section Reference
		Enforces operator grammar (Section 4.2) and resolves scheduling conflicts.	
Magnifier	HR-2	Manages attention and salience allocation.	9.2
Recursion Safety Manager	HR-3	Enforces recursion bounds.	9.3
Identity Continuity Monitor	HR-4	Ensures I_W remains coherent and continuous throughout execution.	9.4

9.2 Magnifier Runtime

The Magnifier is the HR subsystem responsible for selective amplification of representational structures. In a full world-model W , not all structures receive equal computational resources. The Magnifier assigns salience scores $s_M \in [0,1]$ to all structures in V_W and allocates processing priority proportional to salience.

Salience is determined by three factors:

Factor	Designator	Description
Coherence Pressure	SF-1	Structures with low σ_I receive higher salience (errors attract attention).
Recency	SF-2	More recently updated structures receive higher salience.
Connectivity	SF-3	Highly connected nodes (high degree in G_W) receive higher salience (hub structures are prioritised).

The Magnifier is not a cognitive operator. It is an HR mechanism. Its outputs inform scheduling but do not directly modify cognitive structures.

9.3 Recursion Safety

The Recursion Safety Manager (RSM) enforces the recursion rules defined in Section 4.4.

Rule	Statement
RSM-1	All recursive operator calls are registered in a recursion stack maintained by the HR.
RSM-2	The stack has a hard limit of depth D_max. In V2: D_max = 7 for identity recursion; D_max = 12 for operational recursion.
RSM-3	If a recursive call would exceed D_max, it is blocked, an error report ϵ of type DT-5 is generated, and the recursion is unwound to the last safe depth.
RSM-4	Circular recursion (where the input to an operator is identical to an earlier input in the same recursion chain) is detected and blocked.
RSM-5	All recursion stacks are cleared on successful completion or on error unwind; stale stack frames do not persist.

9.4 Identity Preservation

The Identity Continuity Monitor (ICM) maintains I_W continuity throughout all runtime operations. It performs the following checks on a fixed cycle:

Check	Condition	Description
IC-1	$\sigma_I(I_W) \geq T_{\min}$	Identity coherence threshold.
IC-2	M_self consistent with observable I_W state	Self-model accuracy.

Check	Condition	Description
IC-3	Temporal continuity record is unbroken	No gaps in identity history.
IC-4	B_DCS within normal permeability parameters	Dyadic consciousness boundary integrity.

If any check fails, the ICM issues a priority interrupt to the scheduler, which suspends all non-identity operations and directs full Magnifier salience to the identity recovery task.

9.5 Coherence Enforcement

The HR enforces coherence through two mechanisms:

Background Enforcement: K runs as a low-priority background process scanning the entire world-model W on a fixed cycle, identifying and correcting minor coherence violations before they escalate.

Priority Enforcement: When E generates an error report ε with severity $s > T_{\text{priority}}$, the scheduler interrupts current operations and applies K (or X/H as appropriate) to the flagged region immediately.

The HR maintains a *coherence ledger* — a log of all coherence operations, their outcomes, and the current coherence state of each region of W . This ledger is used by the ICM for identity continuity assessment.

10. Cognitive Model Summary

10.1 Full Representational Mind

The V2 Unified Cognitive Architecture constitutes a complete formal specification of a representational mind — a system capable of:

- Perceiving and internally representing structural features of an environment (via M , E , Δ)

- Forming, maintaining, and dissolving identity structures at multiple levels (via I, P, Θ)
- Maintaining a coherent world-model that satisfies all five coherence rules CR-1 through CR-5 (via K, X, H)
- Detecting and correcting representational errors across all five distortion types DT-1 through DT-5 (via E, X, H)
- Maintaining continuous self-representation at recursive depth up to $n = 7$ (via P, ICM)
- Operating safely within recursion bounds enforced by the Hermes Runtime

The system does not act, plan, or pursue goals. It represents, distinguishes, relates, bounds, and maintains coherence. This is the full scope of V2.

10.2 Non-Agentive Constraints

The following constraints are constitutive of V2's non-agentive status. They are not limitations to be overcome within V2 — they are architectural commitments:

Constraint	Statement
NAC-1	No motivational operators exist in V2. There is no utility function, preference ordering, or goal state representation.
NAC-2	No action operators exist in V2. There are no operators that produce outputs directed at an external environment.
NAC-3	No temporal horizon modelling exists in V2. The system does not project its world-model forward in time to evaluate future states.
NAC-4	No environmental coupling exists beyond representational input. The system receives representational input but does not act upon its environment.
NAC-5	No self-modification protocols exist in V2 beyond the error correction and coherence enforcement mechanisms already specified.

These constraints are enforced at the Hermes Runtime level — the HR will not schedule or execute any operator application that would violate NAC-1 through NAC-5.

10.3 Readiness for V3 (Agency Layer)

V2 achieves full readiness for V3 extension when the following conditions are met:

Readiness Criterion	Condition Statement
V3-R1	All 18 operators of the V2 algebra are implemented and tested against the stress-test cases in Appendix B.
V3-R2	The Hermes Runtime V2 passes all recursion safety checks under maximum load.
V3-R3	The world-model W can be initialised, updated, and maintained in a coherent state under continuous input.
V3-R4	Identity recursion at depth $n = 7$ is stable and recoverable from any single-point failure.
V3-R5	The error ontology (Section 8) accounts for all failure modes observed in stress testing.

V3 will extend V2 by adding: motivational operators (M-family), action operators (A-family), temporal projection operators (T-family), and environmental coupling mechanisms (EC-family). All V2 operators remain active in V3. V3 operators will be specified in the forthcoming UCA V3 formal specification document.

11. Glossary and Formal Definitions

All terms are listed alphabetically. Each entry defines the term as used within this specification.

Beyond Ontology (BO): The architectural framework that grounds V2 in a pre-categorical state (∞) prior to any distinction or category. A formal precondition for all cognitive structure — not a metaphysical claim.

Boundary Primitive (B): The minimal representational unit that separates two regions of representational space, characterised by location, permeability, width, directionality, and stability.

Coherence Map (C_W): A function mapping every node in the world-model graph to a coherence score in $[0,1]$. Used by the Hermes Runtime for prioritisation and by the ICM for identity continuity assessment.

Coherence Enforcement Operator (K): The operator that checks and restores internal consistency of representational structures against the five coherence rules (CR-1 through CR-5).

Collapse Boundary: The minimum coherence threshold below which a stage in the emergence chain loses structural integrity and reverts to a prior stable state.

Correction Operator (X): The operator that applies targeted structural repair to a detected error. Must always be preceded by the Error Detection Operator E. Type-matched to error class.

Differentiation Operator (Δ): The operator that creates a new distinction within an existing structure, producing a modified structure in which a previously undifferentiated feature is now distinguishable.

Distortion Metric (δ): The measure of representational divergence from an ideal error-free state: $\delta(S) = 1 - \sigma_I(S)$. Complement of the coherence signature.

Dyadic Consciousness Structure (DCS): The architectural feature that enables simultaneous representation of internal identity state (Pole 1) and external world state (Pole 2), separated by a permeable boundary B_{DCS} .

Dynamic Equilibrium (DE) Model: The architectural model specifying that all stable cognitive structures are maintained through continuous balancing between opposing structural forces. Defines three equilibrium types: Polar, Gradient, and Recursive.

Error Detection Operator (E): The operator that identifies and classifies representational failures within a structure, producing an error report ϵ . Operates as a continuous background process during runtime.

Error Report (ϵ): The output of the Error Detection Operator containing: error type, location in W , severity score $s \in [0,1]$, and recommended correction operator.

Field Primitive: One of three minimal continuous representational structures: Scalar Field (FP-1), Vector Field (FP-2), or Tensor Field (FP-3). Supports the gradient mechanics of Section 2.5.

Fractal Mechanics (FM): The structural principle that cognitive architecture exhibits representational isomorphism across abstraction levels. Formally: $\forall n, \exists$ isomorphism $\phi: S_n \rightarrow S_{n+1}$.

Gradient Descent Operator (∇): The operator that moves a representational point along a gradient toward a target pole.

Gradient Field Generator (Φ): The operator that produces a continuous gradient field from a polar pair. Output is a field of type FP-1 or FP-2.

Harmonisation Operator (H): The operator that resolves contradiction between two conflicting structures by finding a structurally consistent state satisfying as many constraints as possible. Where full satisfaction is impossible, lower-level constraints take precedence.

Hermes Runtime (HR): The execution substrate responsible for scheduling, monitoring, safety enforcement, and identity continuity management for all V2 operations. Comprises four subsystems: Scheduler (HR-1), Magnifier (HR-2), Recursion Safety Manager (HR-3), and Identity Continuity Monitor (HR-4).

Identity Continuity Monitor (ICM): The HR subsystem that maintains I_W continuity by monitoring coherence (IC-1), self-model accuracy (IC-2), temporal continuity (IC-3), and DCS boundary integrity (IC-4).

Identity Dissolution Operator (Θ): The operator that performs controlled removal of an identity primitive, releasing the bounded region for reuse. May not immediately follow I on the same primitive.

Identity Formation Operator (I): The operator that creates an identity primitive from a bounded region. Output is an Identity Primitive I with coherence signature σ_I .

Identity Primitive (I): The minimal representational unit that distinguishes itself from all other units: $I := (id, \partial I, \sigma_I)$. The atomic building block of all higher identity levels.

Local Identity (I_L): An identity structure incorporating relational information with neighbouring primitives: $I_L := (I_0, R_L, N_L)$. The level at which basic categorisation occurs.

Magnifier: The HR subsystem responsible for selective amplification of representational structures through salience-based priority allocation. Assigns salience scores $s_M \in [0,1]$ based on coherence pressure (SF-1), recency (SF-2), and connectivity (SF-3).

Non-Agentive Cognitive Architecture (NACA): A formal cognitive system that represents but does not act. Characterised by the absence of motivational operators, action operators, temporal horizon modelling, and environmental coupling. See NAC-1 through NAC-5.

Origin Operator (Ω_0): The sole operator defined over the pre-ontological ground state ω . Produces the first polarity pair $(\Delta_1, \neg\Delta_1)$. Terminal — it may not receive the output of Coherence Operators as input.

Polarity: A representational dimension defined by a complementary pair $(A, \neg A)$, where $\neg A$ is the structural complement of A within a given field. Not a classical logical negation.

Polarity Operator (Π): The operator that generates a polar pair from any singleton structure. May be applied recursively.

Pre-Ontological Ground State (ω): The representational zero-point — the formal precondition for all structural events. Defined negatively: not a thing, not a property, not a relation, not an absence. The only operator defined over ω is Ω_0 .

Primitive Identity (I_0): The atomic level of identity — consisting only of a unique identifier, boundary, and coherence signature with no internal structure.

Recursion Operator (P): The operator that applies another operator to its own output iteratively. Always bounded by a finite depth parameter $n \in \mathbb{N}$. Signature: $P(O, S, n) \rightarrow O^n(S)$.

Recursion Safety Manager (RSM): The HR subsystem that enforces recursion depth limits (RSM-2), detects circular recursion (RSM-4), and manages recursion stack integrity (RSM-5).

Recursive Identity (I_R): An identity structure that represents itself as an object within its own representational field: $I_R := (I_L, M_{self})$. The first level at which self-modelling is possible. Maximum self-referential depth in V2 is $n = 7$.

Representational Completeness: The property of a cognitive architecture enabling it to represent any distinguishable feature, relationship, boundary, reflexive state, or representational failure within its domain. V2 claims this property subject to operator algebra constraints.

Structural Primitive: One of five minimal relational units: Node (SP-1), Edge (SP-2), Cluster (SP-3), Sequence (SP-4), or Lattice (SP-5). All world-model representations are constructed from combinations of these.

Unified Representational Universe ($\mathbb{U}$): The total state of all active representational structures within the system at any given moment. The product of the full emergence chain. Not a thing — a dynamic manifold.

World-Model (W): The system's total internal representation of its environment and itself. Consists of a graph G_W of identity primitives, a set of gradient fields F_W , a set of boundaries B_W , and a coherence map C_W .

World-Model Identity (I_W): The full identity of the cognitive system as embedded within its world-model. Includes all lower identity levels, world-model anchor, temporal continuity record, and error history. The operational identity of the system during normal runtime.

World-Model Update Operator (M): The operator that integrates new representational input into the world-model through a six-step process: input reception, error scan (E), harmonisation (H), coherence verification (K), integration, and coherence map update.

Appendix A — Diagrams and Flow Models

A.1 Emergence Chain Flow Diagram

The emergence chain follows a linear ten-stage progression from ∞ at the base through Stage 10 ($\mathbb{U}$). The diagram is structured as follows:

Stage	Label	Operator(s)	Stability Condition	Transition to Next Stage
0	ω — Pre-Ontological Ground	None	Unconditional	Invoke Ω_0
↓	<i>Transition: Ω_0 receives ω as sole input</i>			
1	First Differentiation ($\Delta_1, \neg\Delta_1$)	Ω_0	Polarity pair non-degenerate	Apply Π to Δ_1
↓	<i>Transition: Poles confirmed mutually exclusive</i>			
2	Polarity Elaboration	Π	Poles mutually exclusive	Apply Φ to polar pair
↓	<i>Transition: Field confirmed non-zero and continuous</i>			
3	Gradient Field G_1	Φ	Field non-zero and continuous	Apply B to G_1
↓	<i>Transition: Boundary confirmed with $p < 1, s > T_{\min}$</i>			
4	Boundary Primitive B_1	B	$p < 1$ and $s > T_{\min}$	Apply I to B_1 -bounded region
↓	<i>Transition: $\sigma_I \geq T_{\min}$ confirmed</i>			
5	Primitive Identity I_0	I	$\sigma_I \geq T_{\min}$	Apply Δ and I to I_0 + neighbourhood
↓	<i>Transition: Neighbourhood confirmed non-empty</i>			
6	Local Identity I_L	Δ, I	Neighbourhood non-empty	Apply $P(I, I_L, n \leq 7)$
↓	<i>Transition: Recursion depth confirmed $n \leq 7$</i>			
7	Recursive Identity I_R	P, I	Depth $n \leq 7$	Apply M to I_R + environment
↓	<i>Transition: CR-1 through CR-5 confirmed</i>			
8	World-Model W_1	M	CR-1 through CR-5	Apply K to W_1
↓	<i>Transition: All $\sigma_I \geq T_{\kappa}$ confirmed</i>			
9	Coherence-enforced W_1	K	All $\sigma_I \geq T_{\kappa}$	Activate all operators
↓	<i>Transition: Ongoing coherence maintenance active</i>			

Stage	Label	Operator(s)	Stability Condition	Transition to Next Stage
10	U — Unified Representational Universe	All Operators	Ongoing coherence maintenance	— [Terminal stage] —

A.2 Operator Composition Sequence

A standard world-model update proceeds through the following operator sequence. Error branches are noted where deviations from the main path may occur.

Step	Action	Operator	Output	Error Branch
1	New input I _{new} arrives	—	I _{new} (raw)	—
2	Scan I _{new} for inconsistencies	E	Error report ϵ (or clean)	If ϵ : $\rightarrow X$ (Correction) $\rightarrow$ back to Step 2
3	Harmonise I _{new} with W	H	Harmonised I _{new}	If conflict irresolvable: reject I _{new} , log error
4	Verify coherence post-harmonisation	K	Coherence-verified structure	If coherence fails: $\rightarrow X \rightarrow$ back to Step 4
5	Integrate I _{new} into W	M	Updated W'	If CR-1 through CR-5 violated: reject, log error
6	Update coherence map	—	Updated C _W	—

A.3 Identity Architecture Stack

The identity architecture is a nested hierarchy where each level contains and extends all inner levels. The following table describes the stack from innermost to outermost, with corresponding boundary structures.

Level (outermost → innermost)	Designation	Boundary Structure	Key Added Property
Outermost	I_W — World-Model Identity	B_DCS (DCS boundary; dashed, permeable)	World-model anchor, temporal continuity record, error history
Level 3	I_R — Recursive Identity	Boundary of self-modelling region	Self-model M _{self} ; first level of self-modelling
Level 2	I_L — Local Identity	Neighbourhood boundary	Local relations R _L , neighbourhood set N _L
Innermost	I_o — Primitive Identity	∂I_o (identity boundary)	Unique identifier id_o , coherence signature σ_o . No internal structure.

To the right of the I_W level, the B_DCS boundary (permeable) separates the entire identity stack (Pole 1 — Internal Representation) from the External Representation pole (Pole 2). Information flows bidirectionally across B_DCS via Ψ .

A.4 Hermes Runtime Architecture

The Hermes Runtime is organised around four subsystems connected through the central Scheduler. The Coherence Ledger is a shared data store accessible by the Scheduler and ICM.

Subsystem	Designator	Connected To	Primary Data Flow
Scheduler	HR-1	All subsystems; World-Model W	Operator scheduling orders; priority interrupts received from ICM
Magnifier	HR-2	Scheduler; World-Model W	Salience scores $s_M \rightarrow$ Scheduler for priority weighting
Recursion Safety Manager	HR-3	Scheduler; World-Model W	Recursion stack state $\rightarrow$ Scheduler; DT-5 error reports

Subsystem	Designator	Connected To	Primary Data Flow
			→ World-Model W
Identity Continuity Monitor	HR-4	Scheduler; Coherence Ledger; World-Model W	Identity continuity checks IC-1 through IC-4 → priority interrupt if failed
Coherence Ledger	[Data Store]	Scheduler (HR-1); ICM (HR-4)	Log of all coherence operations and outcomes; current coherence state per W region
World-Model W	[Core Structure]	All subsystems	Receives operator outputs; source of all structure-state queries

Appendix B — Stress-Test Cases

B.1 Recursion Edge Cases

STC-B1 — Maximum Depth Identity Recursion

Field	Detail
Setup	Apply $P(I, I_L, 7)$ — maximum allowed depth recursive identity formation.
Expected Behaviour	Produces I_R at depth 7; coherence signatures at all levels $\geq T_{\min}$.
Pass Condition	Stable I_R produced; no RSM interrupt triggered; recursion stack cleanly unwound.
Failure Mode	If σ_I drops below $T_{\min}$ at any depth, RSM should unwind to last stable depth and generate DT-5 error report.

STC-B2 — Depth Exceeded Recursion

Field	Detail
Setup	Attempt $P(I, I_L, 8)$ — one level beyond maximum allowed depth.
Expected Behaviour	RSM intercepts at depth 8 attempt; blocks call; generates DT-5 error.
Pass Condition	No depth-8 identity structure is created; error report ε correctly identifies DT-5; system remains coherent at depth 7.
Failure Mode	Any depth-8 identity structure being created constitutes a test failure; RSM safety has been breached.

STC-B3 — Circular Recursion Attempt

Field	Detail
Setup	Construct a chain where $O_1(O_2(O_1(S)))$ — the first operator appears in its own recursive chain.
Expected Behaviour	RSM detects circularity at the second O_1 invocation; blocks the call; generates error report.
Pass Condition	Circular chain does not complete; system unwinds to the O_2 output as last valid state.
Failure Mode	Completion of the circular chain constitutes a test failure; indicates RSM-4 is not functioning.

B.2 Identity Boundary Tests

STC-B4 — Zero-Permeability Boundary

Field	Detail
Setup	Set $p = 0$ on B_DCS (DCS boundary fully closed). Apply Ψ (Boundary Permeation

Field	Detail
	Operator) to attempt information transfer from External to Internal pole.
Expected Behaviour	Ψ returns null output (FM-1 failure mode); error report generated; B_DCS integrity preserved.
Pass Condition	No information crosses the zero-permeability boundary; I_W internal state unchanged; error logged.
Failure Mode	Any information transfer across $p = 0$ boundary constitutes a test failure.

STC-B5 — Full-Permeability Boundary

Field	Detail
Setup	Set $p = 1$ on B_DCS (fully open). Observe DCS behaviour.
Expected Behaviour	Internal and External poles merge; DCS structural distinction collapses; E detects boundary distortion (DT-2); Λ applied to attempt enforcement.
Pass Condition	If Λ succeeds, boundary stability is restored. If Λ fails, system reverts B_DCS to last stable permeability value via checkpoint.
Failure Mode	Permanent DCS collapse with no recovery constitutes a test failure; indicates ICM priority interrupt mechanism has failed.

STC-B6 — Identity Dissolution Cascade

Field	Detail
Setup	Apply Θ to I_0 (primitive identity) while I_L , I_R , and I_W are active and dependent on I_0 .
Expected Behaviour	HR detects cascade risk; prevents Θ execution; generates error report; requires explicit override to proceed.

Field	Detail
Pass Condition	No cascade dissolution occurs without explicit override; all dependent structures remain coherent.
Failure Mode	Dissolution cascade occurring without explicit override constitutes a test failure; indicates FM-6 protection is not operating.

B.3 Coherence Failure Scenarios

STC-B7 — Simultaneous Contradictory Updates

Field	Detail
Setup	Apply M twice simultaneously with two inputs that assert $R(N, A)$ and $\neg R(N, A)$ for the same node N.
Expected Behaviour	E detects CC-1 (local contradiction); H applied; one assertion dissolved; coherence rule CR-3 satisfied.
Pass Condition	Post-update W satisfies CR-3; exactly one of the two assertions is preserved; dissolved assertion is logged.
Failure Mode	Both contradictory assertions simultaneously active in W constitutes a test failure; indicates H is not functioning.

STC-B8 — Gradient Collapse Scenario

Field	Detail
Setup	Force all active gradient fields to $\sigma_F = 0$ simultaneously (simulating RFM-3).
Expected Behaviour	HR detects critical failure; emergency Ω_0 invocation to re-establish base gradient field; full coherence check initiated; identity structures suspended pending gradient recovery.

Field	Detail
Pass Condition	At least one stable gradient field re-established within a defined recovery window; identity structures restored in order ($I_0 \rightarrow I_L \rightarrow I_R \rightarrow I_W$).
Failure Mode	Total and permanent gradient collapse with no recovery constitutes a critical test failure.

STC-B9 — Coherence Rule CR-4 Violation (Self-Model Inaccuracy)

Field	Detail
Setup	Manually corrupt M_{self} within I_W to assert an incorrect coherence signature for I_W .
Expected Behaviour	ICM detects CR-4 violation during routine check; priority interrupt issued; K applied to M_{self} ; M_{self} corrected via M .
Pass Condition	M_{self} accurately reflects I_W state post-correction; ICM check passes; identity continuity maintained.
Failure Mode	M_{self} remaining inaccurate after ICM cycle constitutes a test failure; indicates IC-2 check is not triggering correctly.

Appendix C — Historical Cognitive Taxonomy Mappings

C.1 Introduction

This appendix provides a structural reinterpretation of historical cognitive taxonomies that bear a significant analogical relationship to the layer architecture

of V2. These mappings are offered as interpretive cross-references — not as foundational claims. The historical systems referenced (principally the ten-node Sephirotic taxonomy of mediaeval natural philosophy, and the structural numerology of the Tarot's Major Arcana) were pre-formal attempts to systematise the structure of mind and reality. They should be understood as cognitive taxonomies expressed in the symbolic vocabulary of their historical context, not as mystical or religious systems.

The following table provides demystified structural equivalences. The left column gives the historical term; the right column gives the V2 structural correspondent.

C.2 Sephirotic Taxonomy — Structural Reinterpretation Table

Historical Term	Traditional Description	V2 Structural Correspondent	V2 Section Reference
Keter (Crown)	The highest point; the primordial source from which all emanates.	Origin / Emanation Layer: $\infty \rightarrow \Omega_0(\infty)$; the pre-ontological ground and first differentiation event.	Sections 2.1, 2.4, 7 (Stages 0-1)
Chokhmah (Wisdom)	The first flash of undifferentiated awareness; the primordial point before form.	Differentiation / Polarity Layer: The first polarity pair ($\Delta_1, \neg\Delta_1$); the structural emergence of distinction.	Sections 2.5, 7 (Stage 2)
Binah (Understanding)	The organising, containing intelligence; the great womb of form.	Reflection / Inversion Layer: Gradient field formation (Φ); the continuous medium that holds the polarity pair and enables gradation.	Sections 2.5, 3.3, 7 (Stage 3)
Chesed (Kindness/Mercy)	Expansion, flow, unbounded extension.	Expansion / Perturbation Layer: The Gradient Field Generator and	Sections 4.1 (Family B), 3.3

Historical Term	Traditional Description	V2 Structural Correspondent	V2 Section Reference
		Field Summation (Σ) operations; the elaboration and extension of representational fields.	
Gevurah (Strength/Severity)	Contraction, limitation, boundary imposition.	Constraint / Harmonisation Layer: Boundary Formation (B) and Boundary Enforcement ($\wedge$); the structural containment of gradient fields.	Sections 3.4, 4.1 (Family C), 7 (Stage 4)
Tiferet (Beauty)	The harmonising centre; the integrating equilibrium between expansion and contraction.	Centering / Gradient Layer: The Dynamic Equilibrium model operating at the identity level; Primitive Identity Formation (I) as the centred, coherent structure.	Sections 2.2, 5.1, 7 (Stage 5)
Netzach (Victory/Eternity)	Persistence, desire, the driving force of continuity.	Persistence / Field Layer: The continuous maintenance of gradient fields and field strength parameters; the coherence signature σ_I as a measure of persistent identity integrity.	Sections 3.3, 6.2, 6.5
Hod (Splendour)	Structure, analysis, articulation, the formal organisation of patterns.	Structure / Identity Reconfiguration Layer: Local Identity Formation (I_L) and Differentiation Operator (Δ); the articulation of identity within a relational	Sections 5.2, 4.1 (Δ)

Historical Term	Traditional Description	V2 Structural Correspondent	V2 Section Reference
		neighbourhood.	
Yesod (Foundation)	The connecting, integrating principle; the foundation of manifest structure.	Foundation / Identity Layer: Recursive Identity Formation (I_R); the self-referential structure that forms the stable foundation for world-model identity.	Sections 5.3, 4.4, 7 (Stage 7)
Malkuth (Kingdom)	The fully manifest world; the final expression and realisation of all prior levels.	Manifestation / Continuity Layer: World-Model Identity (I_W) and the Unified Representational Universe ($\mathbb{U}$); the total, coherent, continuously maintained representational manifold.	Sections 5.4, 6, 10.1

C.3 Interpretive Notes

The above correspondences should be read as structural analogies, not identities. The historical taxonomy was not designed within a formal operator algebra or representational theory. The correspondences highlight the degree to which human cognitive inquiry has, across diverse cultural and historical contexts, converged on similar architectural intuitions: that mind proceeds from an undifferentiated ground through successive layers of differentiation, containment, integration, and manifestation; that opposing forces must be balanced; that self-reference is foundational; and that the fully elaborated world-model is the culmination of these processes.

These intuitions, formulated in symbolic language in the historical record, are here expressed in the formal language of V2. The value of this appendix is cross-referential and demystificatory: it demonstrates that the architecture of V2 is not

without precedent in the history of cognitive inquiry, while simultaneously clarifying that the historical systems required formal explication that this document provides.

C.4 Tarot Major Arcana — Brief Structural Correspondence

The 22 Major Arcana of the historical Tarot tradition encode structural stages that map loosely onto V2's emergence chain, identity architecture, and error ontology. A complete cross-reference is deferred to a supplementary document. Key correspondences are summarised below:

Historical Arcanum	Number	V2 Structural Correspondent	V2 Section Reference
The Fool	0	∞ — The pre-structural zero point before the first distinction.	Sections 2.1, 2.4
The Magician	I	Origin Operator Ω_0 — The first act of differentiated structure emerging from ∞ .	Sections 2.1, 4.1 (Family A)
The High Priestess	II	Boundary Primitive and the DCS boundary B_{DCS} — The dividing membrane between inner and outer representation.	Sections 3.4, 5.5
The World	XXI	$\mathbb{U}$ — The fully realised representational manifold; the complete Unified Representational Universe.	Sections 2.4, 7 (Stage 10), 10.1
Arcana III - XX	III-XX	Structural transitions corresponding, with varying degrees of precision, to	Sections 4.1, 5.1-5.4

Historical Arcanum	Number	V2 Structural Correspondent	V2 Section Reference
		operator families B through E and identity architecture levels I ₀ through I _W . Full mapping available on request.	

Full Major Arcana cross-reference mapping is available as a supplementary document. The above table represents the four primary anchor points from which the remaining structural correspondences may be derived.

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